L1.3: Systems of Linear Equations

Transcript Language

English

Welcome to the maths 2 component of the online B.Sc degree in data science. In today's video

we are going to look at systems of linear equations. So, what are the contents of this

video? So, let us see what is there, what is the system of linear equations? What is

its relation with matrices? And finally, how many solutions can it have? So, these are the three things we are going to see today.

Let us begin with an example, so in this example we have 3 buyers and we have 3 items, so buyer

A buys rice 8 kg s of rice, 8 kg s of dal and 4 litres of oil, buyer B buys 12 kg s

of rice, 5 kg s of dal and 7 litres of oil and buyer C buys 3 kg s of rice, 2 kg s of

dal and 5 litres of oil.

So, suppose these buyers paid some amount of money to the shopkeeper, namely A paid

rupees 1960, B paid rupees 2215 and C paid rupees 1135, so I have this data. Now, we

want to find the price of each item using this data. So, how do I write this as a system

of linear equations? So, we will assume that the price of rice is rupees x per kg, we will

assume that the price of dal is rupees y per kg and we will assume that the price of oil

is rupees z per litre.

So, then what do we get for the first buyer? We get that 8 times x, because he bought he or she bought 8 kg of rice, so 8 times x plus 8 times y 8 kg s of dal plus 4 times z is

1960. Similarly, we get the next two equations. So, this is what a system of linear equations

looks like and now from this data we would like to solve this system and get what is

the price of rice, dal and oil.

So, I am going to quickly solve this. So, I think many of you may have solved such things

in school, so how do we solve this? So, maybe we can take the first equation and divide

it by 2, so if you do that, we get 4x plus 4y plus 2z is 980, this the first equation

divided by 2 and then we will multiply this new equation by 3, so that I get a 12x, so

12x plus 12y plus 6z which is 2940.

Then I take this equation and subtract out the second equation over here, I will subtract

out the second equation over here, so if I do that, what do I get? So, the 12x is subtracted

out and what we get is 7y minus z is 725. So, now we have an equation in terms of y and z, now similarly for the third equation what we can do is we can use the third equation

and multiply the third equation by 4, so that we get a 12x.

So, multiplying the third equation by 4, we will get 12x plus 8y plus 20z is 4 times 1135

and then we will use this equation and subtract out the second equation, so what we get is

another equation in terms of y and z, so the equation we get in terms of y and z is 3y

plus 13z is 2325, so I will encourage it to do this. So, now we have two equations with

two unknowns, namely y and z, so let us try and solve this.

So, well we can actually solve this by putting z equals 7y minus 725 and substitute in the

second one, so if you substitute in the second one we get 3y plus 13 times 7y minus 725 is

2325. So, if you simplify this what you will get is 94 times y is 11750, so then y is 11750

divided by 94 which is 125. So, we have computed the price of dal, it is rupees 125 per kg,

so we will substitute y is 125 in this equation over here and we will get z, so z is from here we will get that z is 150.

And then now we know y and z, so we will put them in some suitable equation and we can

get x. So, x is going to be 45, so I will encourage you to check these. So, what was

the point of this? The point of this was we had a real life problem in involving some

costs and so on, so we have three equations.

And then we use some unknowns to denote the things that we wanted to compute and then

we had three linear equations in those three unknowns and then we solve the system of linear

equations. And we computed that x is 45, y is 125 and z is 150 which means the price

of rice is rupees 45 per kg, the price of dal is rupees 125 per kg and the price of

oil is rupees 150 per litre. So, this I hope this convinces you that solving a system of

linear equations is something useful.

So, what is the system of linear equations? So, before that we have to ask, what is a linear equation? So, a linear equation is something of the form a1x1 plus a2x2 plus

anxn is b, this is what a linear equation looks like, this is one linear equation. So,

what are the x1 x2 xn these are variables or unknowns, these are the things that we

want to compute, this was like the x y z in our previous example.

And then what are the a1 a2 an s? These are what are called coefficients, these are going

to be real numbers, these are things we will assume, these are numbers will assume that

we know. And then b is also a real number. So, here is an example, so 2x plus 3y plus

5z is minus 9 and here x y z and are variables and the 2, 3, 5, are coefficients.

So, what is the system of linear equations? So, a system of linear equations, so that

means you have several linear equations, we saw what is one linear equation, so it is

something of the form a1x1 plus a2x2 plus anxn is b, instead now we have many such equations

but in the same set of unknowns, so x1 x2 xn remains the same but the coefficients and the constant at the end change.

So, here is an example, so we have 3x plus 2y plus z is 6, x minus half y plus two third

z is 7 6th and 4x plus 6y minus 10z is 0. So, I will encourage you to solve this equation

similar to how we did it in the first example. So, if you do that you should get the answer

x is 1, y is 1 and then 1 is 1. So, what are the unknowns or the variables here?

The x, y and z are the unknowns of the variable, so traditionally you represent if you have

one unknown it x, if you have two unknowns x and y, if you have three unknown x, y and

z and if you have four or more unknowns you take x1 x2 x3 x4 and up till as many unknowns

as you have. So, what is the solution to this system of linear equations?

That means if you put x equals 1, y equals 1 and z equals 1 in this system here, then

these equations are solved. So, let us check that, so 3 times 1 plus 2 times 1 plus 1 is

indeed 6, 1 minus half times 1 plus 2 3rd times 1, so let us 1 minus so that is 1 minus so that is half plus 2 3rd which is indeed 7 by 6.

And then similarly you have 4 times 1 plus 6 times 1 minus 10 times 1 is 0. So, I hope

it means you understand what it means, for a system of linear equations to have a set

of solutions, so when you put x, y and z to be some particular numbers then these equations

should be satisfied, that means these equality should be true, that means we say we have

got solution.

So, what is the general system of linear equations? So, you have m linear equations, that means

you have m equations and you have n unknowns, so n unknowns mean you have x1 x2 xn, then

how do we write these? We write this as a11 x1 plus a12 x2 plus a1n xn is b1, a21 x1 plus

a22 x2 plus a2 nx2 is b2 and all the way up to am1 x1 plus am2 x2 plus amn xn, is bm.

So, if you want the ith equation, then this is ai1 x1 plus ai2 x2 plus ain xn is bn this

is the ith equation in this system.

So, what is the connection with matrices? So, this is really important part of this

video, so the system of linear equations is equivalent to a matrix equation of the form

a times x is b, so in a previous video we have seen what it means to multiply matrices.

So, here Ax and b are matrices, so A is an m by n matrix and what are the entries of

this matrix? They are exactly the Aij s that we saw in our previous slide.

So, this matrix A consists of the Aij s, x is a column vector with n entries, what are

the entries? x1 x2 xn exactly or unknowns and b is the column vector b1 b2 bm, so this

is A that from our previous slide these are the coefficient, so this A is called the coefficient

matrix. What is x? x is a column matrix, so it that means it consist of one column, so

it is the column x1 x2 xn, it is a matrix with n rows and 1 column, it is a column matrix

consisting of the unknowns.

And b is the set of constants, so it is a again a column matrix consisting on the constants.

So, this is a m by 1 matrix, this is n by 1 and this is m by n. And now if you multiply

these first of all let us ask does it make sense indeed this is m by n and this is n

by 1, so it makes sense to multiply these and whatever the product is it will be of

size m by 1.

So, we are saying that a product is exactly the left-hand side of the system of linear

equations that we had on the previous slide and on the right hand side we will have these

constant b1 b2 bm. So, the previous system can be expressed in the form ax equals b,

so the important part here is that any system of linear equations can be expressed in terms

of matrices as a matrix by matrix multiplication. Let us, look at some examples.

So, let us look at this example like we had before, 3x plus 2y plus z is 6, x minus half

y plus 2 3rd z is 7 6, 4x plus 6y minus 10z is 0, so how do we represent this in terms

of matrices? So, in terms of matrices this is the coefficient matrix, so we collect together all the coefficients and put them in the correct places. So, the first row is the 2 1, that

is what occurs as a coefficient of in the first equation.

The second row is 1 minus half 2 3rd s that is what occurs in the second equation and

the third row is 4 6 minus 10, that is what occurs in the third equation. What is x? x

is x y z, the unknowns or the variable. And what is b? b is the set of constants on the

right, so 6 7 6 and 0 and I will encourage you to multiply A and x.

So, multiply A by x and see that you get exactly this thing over here as a column matrix, so

you get exactly this as your matrix. This is A times x and this thing on the right is

b. And what are the system of linear equations saying? Saying Ax equals b. So, we have rewritten

this example as a in terms of matrix multiplication. Let us look at our first example.

So, this was 8x plus 8y plus 4z is 1960, 12x plus 5y plus 7z is 2215, 3x plus 2y plus 5z

is 1135. How do we write this in terms of matrices? So, the coefficient matrix here is going to be 8 8 4, 12 5 7, and 3 2 5, the matrix x which is a column matrix, that will

be x y z and the matrix of coefficients sorry of constants is going to be 1960 2215 1135.

And once again, I will encourage you to check that Ax is equal to be exactly captures the

systems of linear equations upstairs.

Now, I will point out one small strangeness in this entire notation you will see that

the matrix here is called x, whereas you also have a variable called x, so be very careful

to distinguish between these two, so which one is your variable and which one is your

matrix, so when I when we are writing this matrix, this are 3 rows and 1 column, so is

the matrix this is a column matrix of size 3, so 3 rows and 1 column.

And inside it is the one one entry is x which is exactly our variable inside in our equations.

So, this is an unfortunate fact that there is a clash of notations, but that is how it

is and it with some practice you will get used to it. So, we have seen two examples of how the connection between system of linear equation and how to represent them in matrix

terms.

So, now let us talk about solutions, solutions to a linear system. So, we saw I explicitly

found out the solutions for the first example and I asked you to check that x is 1, y is

1, z is 1 the solution for the second example, so in general what can we say about solutions

of a linear system? So, it turns out that there are only three possibilities either

a system has infinitely many solutions or a system has a single unique solution or a

system has no solution.

So, these are the only three possibilities. So, it cannot happen that a system of linear

equation over the real numbers has only three solutions, this is not possible. If it has

three solutions it will have infinitely many solutions, it cannot have exactly three solutions,

so it can have either 0 solutions, 1 solution or infinitely many solutions.

So, let us see examples of where this comes from. So, suppose we have to buyers A and

B and buyer A buys 2 kg s of rice and 1 kg of dal, buyer B buys 4 kg s of rice and 2

kg s of dal.

And they pay 215 rupees and 430 rupees respectively. So, again as in the first example, we want

to find out the price of rice and the price of dal, so we put this into the form of linear

equations. So, if you follow the first example you will quickly see that this is 2 times

x plus y is 215, 4 times x plus 2y is 430. And there are infinitely many x and y satisfying

both the equations. So, I will encourage you to find out why that is the case.

Let me give you one example, I mean let me at least show you two examples, two solutions

and you can find as many more as you want, one solution is x is 0 and y is 215, that

is one solution. So, check that indeed this works. Another solution is x is 215 by 2 so

that is 107.5 and y is 0. So, this is another solution. And I will encourage you to find more solutions. So, what is happening here in terms of whether some geometry involved?

So, let us draw this picture. So, the idea is that if you draw remember that if you have

the equations correspond to straight lines, yes, this is something you have seen in math

1, so you have 2x plus y is 215 and 4x plus 2y is 430 and both of them give you the same

straight line, so any point on this line is going to satisfy is going to be a solution.

So, every point on this line is a solution point. So, now you can find as many solutions

as you want. So, this is the geometry behind why it has infinitely many solutions.

Let us, do an example where it has no solutions. So, again we have the same situation in the

A and B go to the shop by same amount of rice and dal as the previous limit, so 4 kg s of

rice and 2 kg s of dal and 2 kg s of rice and 1 kg of dal respectively, but this time

the seller gives a discount to B, the seller is in a happy mood the shopkeeper and they

give a discount to B.

So, A pays 215 rupees and B paid 400 rupees, in the previous example B had paid 430, this

time they got a discount of 400 rupees. Now, after returning home, they tried to solve

this system again put x and y to be the price of rice and dal respectively, so we get 2x

plus y is 215, 4x plus 2y is 400, let us try and solve this system of linear equations.

So, if you try and solve this, so let us take the first equation and double it.

So if you double the first equation you get 4x plus 2y is 430, on the other hand the second

equation is 4x plus 2y is 400, so what does that mean? That means 400 is equal to 430,

so there is something really really wrong, this is this is something is going very wrong.

So, of course, we know that 400 is not 430 that means there is no x and y for which this

can be solved, so there is no solution here. What goes wrong?

So, the reason there is no solution here is that if you draw the lines in this case, they

are parallel lines, so they would not intersect at all. And the intersection is exactly what will give you the solution. So, in the previous case you had both the equations at the same

line, so it was like you are two lines, which were the same, so the intersected in infinitely

many points. So, here you have two lines which are parallel they do not intersect at all

and so there is no solution.

Finally, let us do the example the last example where you have a unique solution. So, in this

case buyer A buys 2 kg s of rice and 1 kg of dal and buyer B buys rice 3 kg s of rice

and 1 kg of dal.

And buyer A pays 215 rupees buyer B pays 260 rupees, let us write down the equation so

it is 2x plus y is 253, 3x plus y is 260.

And I will encourage you to solve that equation. The point here is you have two lines, 2x plus

y is, maybe let us go back 2x plus y is 215 and 3x plus y is 260, so it is clear from

here that x is 45 and once we know that x is 45 we can compute that y is, so 215 minus

90, so that is 125. So, this is like in our first example, so the price of rice is 45

rupees per kg and the price of dal is 215 rupees per kg.

And you can check here that that is exactly where these intersect, so this is on the x

axis will intersects at 45 and on the y axis, 125 and that is exactly the point of intersection.

So, that is why there is a unique solution. So, you can draw these lines and wherever

they intersect is where you have an equation, you have a solution. So, let us recall quickly

what we study today.

We have studied what is the system of linear equations; we have study now to write it in

terms of matrices, particularly using matrix multiplication with unknowns where x was a

column matrix of variables. And then we have seen that what is the solution to system of

linear equations and then we saw that there are three possible cases either a system has

no solution or it has one unique solution or it has infinitely many solutions. That is all for today. Thank you.